

COEP Technological University (COEP Tech)

Second Year CIVIL Engineering (2024 Pattern)

OPEN ELECTIVE – I

INTERNATIONAL STANDARDS

Syllabus Drafted By-

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COURSE OBJECTIVES

- Provide students with deep insights into engineering standardization—its history, structure, types, and applications, emphasizing major international standards (ISO, IEC, EN, ASME, BIS, etc.), how they're developed, and why they matter in product design and quality assurance.

COURSE OUTCOMES

On successful completion of this course, students will be able to:

- CO1: Understand and Compare between Standards and specifications, Standardization meaning & Standardization Bodies with their Frameworks
- CO2: Identify core international standardization organizations and their roles.
- CO3: Analyze how standards impact global trade, product design, and regulatory compliance.
- CO4: Apply knowledge of standards in real-world case studies and quality management systems

UNIT I

INTRODUCTION TO STANDARDIZATION & STANDARDIZATION BODIES WITH THEIR FRAMEWORKS

- Introduction to Standards and Standardization: Definition and purpose of standards, Types of standards: voluntary vs. mandatory, History and evolution of standardization, Importance in trade and industry, Definitions: Standards vs. specifications, Importance: Safety, quality. ISO: structure, committees, e.g., ISO/TCs, IEC, EN (European standards), ASME, AMISA, BIS (India) International Classification for Standards & International Standardization Bodies : ISO (International Organization for Standardization), IEC (International Electrotechnical Commission), ITU (International Telecommunication Union), ASTM International, IEEE, etc., Role of WTO and TBT Agreement

UNIT II

ISO STANDARDS OVERVIEW & CORE INTERNATIONAL STANDARDS

- ISO 9000 series (Quality Management Systems), ISO 14000 series (Environmental Management Systems), ISO/IEC 27001 (Information Security), ISO 45001 (Occupational Health and Safety), Case studies and applications, ISO 128 – Technical drawing conventions , ISO 10303 (STEP) – Product data modelling/exchange , EN standards (EN ISO series): e.g., mechanical, medical devices , ASME B5 – machine tool tolerances and components, BIS role in harmonizing Indian standards with global benchmarks

UNIT III

STANDARDS DEVELOPMENT PROCESS OF STANDARDIZATION

- Stages of standard development (proposal, drafting, approval), Stakeholders in the standardization process, National participation and harmonization with global standards, Technical Committees and Working Groups, Mapping standards in subjects: e.g., technical drawings (ISO 128 in Engineering Graphics), CAD/CAM (ISO 10303 in CAD), IS Codes in Civil Engineering, Laboratory applications: standard-guided experimentation and measurement. From proposal to publication: committee roles, drafts, ballots, Revision cycles—case studies: e.g., ISO 128's updates

UNIT NO : IV

APPLICATION IN INDUSTRY & COMPLIANCE

- Implementing standards in product design, manufacturing, documentation, Compliance tools: certificates, audits, conformity assessment. Use of standards in manufacturing, IT, health, energy, etc., Quality assurance and standard compliance, Risk management and legal implications, Emerging trends in standardization (digital standards, sustainability, etc.)

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INTRODUCTION TO STANDARDIZATION & STANDARDIZATION BODIES WITH THEIR FRAMEWORKS



HISTORY OF STANDARDISATION

- <https://youtu.be/w7npehPpU74>

STANDARDS:

An unseen force behind technological innovation and adoption.

You can't see standards, but they are at work behind the scenes everyday.

From device to device, network to network, across borders and around the world, almost all products, services and technologies in use today are created, connected or enabled in some way by standards.



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I.Introduction to Standards and Standardization:

- Standards are published documents that establish **technical specifications and procedures designed** to maximize the reliability of the materials, products, methods, and/or services people use every day.
- A standard can be as an **agreed-upon norm** used by people, industry, and government that **outlines the best way to complete a task** – whether it's about developing a product, providing a service, controlling a process, or interacting with the world.

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Definition of Standards

- A **standard** is a documented agreement containing **technical specifications** or other precise criteria to be used consistently as rules, guidelines, or definitions.
- Examples:
 - Traffic signals (red means stop, green means go).
 - The USB-C charging port for many devices.
 - industry-specific guidelines.
 - ISO 9001 for quality management in manufacturing (e.g., Toyota)
 - **IEEE 802.15.1™** Bluetooth standard.



HOW ARE STANDARDS CREATED?

IEEE (Institute of Electrical and Electronics Engineers) Standards Association (IEEE SA)

- which brings together a broad range of people and organizations from different technical and geographic points of origin into “**working groups**” that are open to anyone to participate.
- Many standards are used by multiple industries, such as **health care, automotive, and consumer electronics.**
- That’s why IEEE SA welcomes stakeholders from each vertical market that could use a standard under development.

HOW ARE STANDARDS CREATED?

- These volunteer working groups create a **collaborative environment** for developing standards.
- Suppose members of a working group come together because they agree that **there needs to be a new way for devices to transfer moderate amounts of data over short distances, with very low power consumption to maximize battery life.**
- That's how the **IEEE 802.15.1™** Bluetooth standard was developed — and why your earbuds, smartphone, and smartwatch all can communicate seamlessly with one another.

IMPORTANCE OF STANDARDS:

- Standards form the **fundamental building blocks** for product development by **establishing consistent protocols** that can be **universally understood and adopted**.
- This helps to **simplifies product development, and speeds time-to-market**.
- Standards also make it easier to **understand and compare competing products**.
- As **standards are globally adopted** and applied in many markets, they also fuel **international trade**.

IMPORTANCE OF STANDARDS:

- For example,
- A **standard provides detailed information** that **saves manufacturers enormous amounts of time and money** when developing products.
- They **don't have to create that framework from scratch**, troubleshoot and refine it, and convince other manufacturers to implement it.

IMPORTANCE OF STANDARDS: **FOR BUSINESSES**

1. Improved Quality and Safety :

Standards help businesses **improve the quality and safety of their products and services**, leading to **increased customer satisfaction** and reduced risk of product recalls or liability issues.

2. Reduce cost :

Standards can **streamline processes, reduce waste, and improve efficiency**, ultimately lowering production costs



IMPORTANCE OF STANDARDS: FOR BUSINESSES

3. Enhanced Market Access :

Standards facilitate access to new markets by ensuring products and services meet the necessary requirements for **international trade**.

4. Faster time to market :

By providing a **defined framework**, standards can help **businesses accelerate** the development and **launch of new products and services**.



IMPORTANCE OF STANDARDS: FOR BUSINESSES

5. Increased Competitiveness :

Standards can differentiate businesses by demonstrating their **commitment to quality and safety**, leading to a competitive advantage.

IMPORTANCE OF STANDARDS: FOR CONSUMERS

1. Safety and Reliability :

Standards ensure products and services are **safe and reliable**, providing consumers with **confidence in their purchases**.

2. Quality and Reliability:

Standards define performance characteristics, ensuring **products function as expected and consistently meet specified criteria**. This helps consumers avoid substandard or unreliable goods.



IMPORTANCE OF STANDARDS: FOR CONSUMERS

- **Informed Choices:**

Standards facilitate **comparison between different products and services**, allowing consumers to make informed decisions based on their needs and preferences.

- **Transparency and Information:**

Standards often **require clear labelling and information about products**, enabling consumers to understand what they are purchasing and how to use it safely.



IMPORTANCE OF STANDARDS: FOR CONSUMERS

- **Fairness and Competition:**

Standards promote fair competition by ensuring that all **businesses adhere to the same basic requirements**, preventing unfair practices and deceptive claims.

- **Environmental Protection:**

Standards can **address environmental concerns** by setting requirements for **resource efficiency, waste reduction, and emissions control, supporting sustainable consumption**.



IMPORTANCE OF STANDARDS: FOR CONSUMERS

- **Building Trust:**

Consistent adherence to **standards builds consumer trust and confidence in products, services**, and the businesses that provide them.

- **Consumer Protection Laws:**

Standards often form the basis for consumer protection laws and regulations, **providing a framework for addressing complaints and seeking redress for harm caused by substandard products or services.**



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2. Purpose of Standards:

The primary purposes of standards are:

- I. **Enhancing quality** : Standards define **requirements, specifications, and guidelines** that ensure products, processes, and services are fit for their intended purpose.
 - This helps maintain a **consistent level of quality, reducing variation and improving reliability.**

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2. Building Trust and Confidence:

Standards help **build trust and confidence among consumers** by providing assurance that products and services meet **certain quality and safety requirements**.

This **can increase sales and encourage the adoption of new technologies**.



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3. Promoting Safety: Standards play a **crucial role in enhancing safety** by establishing **benchmarks for safe design, operation, and usage**.

They help prevent accidents, injuries, and damage to property, particularly in areas like electrical systems, food handling, and construction.

4. Facilitating Interoperability: Standards **enable different devices, systems, and components to work together seamlessly**. This is particularly important in areas like software, telecommunications, and transportation, where compatibility is essential.



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5. Streamlining Processes: By providing clear guidelines, standards help simplify and standardize processes, reducing complexity and improving efficiency.

This can lead to cost savings, reduced waste, and faster production times.

6. Supporting Trade: Standards create a common language and framework for international trade, making it easier for businesses to operate across borders.

They reduce trade barriers and promote fair competition by ensuring that products and services meet the same requirements.



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Types of Standards

Voluntary Standards : guidelines and requirements that organizations choose to adopt, **rather than being mandated by law.**

- Developed by organizations through consent.
- **Adoption is optional** but often becomes industry norms.
- Examples: ISO 9001 (Quality Management), IEEE standards.

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- **Characteristics of voluntary standards:**
 - Non-mandatory
 - Developed by stakeholders
 - Promote best practices
 - Widely recognized and accepted
 - Can become mandatory

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- **Benefits of voluntary standard**

- Improved quality and safety
- Increased efficiency and productivity
- Enhanced credibility and reputation
- Access to new markets
- Reduced regulatory burden

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Types of Standards

Mandatory Standards: legally required rules and regulations set by governments or international bodies, ensuring a minimum level of safety, performance, or other specified characteristics for products or services.

- Enforced by law or regulation.
- Non-compliance leads to legal consequences.
- Examples: Building codes, food safety regulations

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- **Characteristics of Mandatory standards**
- **Legally Binding**
- **Government or International Body Enforcement**
- **Safety and Performance Requirements**
- **Consequences of Non-Compliance**

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4. History and Evolution of Standardization:

- **Ancient civilizations** used primitive standards for weights and measures.
- **Industrial Revolution:** Need for uniformity in manufacturing parts led to formal standardization.
- **20th century:** Formation of international standardization bodies (ISO in 1947, IEC in 1906).
- **21st century:** Focus on global harmonization and digital standards.

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Importance of Standards in Trade and Industry

- **Benefits**
- Promote **global trade** by removing technical barriers
- Ensure **interoperability** of systems and devices
- Guarantee **product safety and quality**
- **Reduce costs** through uniform production and economies of scale
- Facilitate **innovation** by providing common platforms

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6. Key Definitions

Term	Definition
Standard	Documented rule or guideline developed by consensus.
Specification	Detailed description of requirements for materials or processes.

Key Differences

- **Standards** are consensus-based and often broad.
- **Specifications** are detailed technical requirements, often used in contracts.

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ISO (International Organization for Standardization)

- The International Organization for Standardization (ISO) is a **global body** that develops and publishes International Standards for various sectors.
- It's a network of national standards bodies from over **160 countries**.
- ISO's mission is to promote standardization globally, **facilitating the exchange of goods and services and fostering cooperation in technical, scientific, and economic fields**.



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ISO (International Organization for Standardization)

How it works : ISO brings together experts from member countries to develop standards through a consensus-based process.

What it covers : ISO develops standards for a **wide range of areas**, including **product specifications, processes, services, and systems**.

Impact : ISO standards **help ensure quality, safety, and efficiency**, making it **easier for businesses** to operate across borders and for consumers to trust products and services.

Examples: ISO 9001 ([quality management](#)) and ISO 14001 ([environmental management](#)).



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IEC (International Electrotechnical Commission)

The world's leading organization that **prepares and publishes international standards** for all electrical, electronic and related technologies.

International standards **provide the technical rules** that **ensure the safety and performance of products** and help reduce barriers to trade.

They provide the basis for conformance **testing and allow electrical and electronic devices to enter markets around the world.**



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IEC certification -

- **Importance of IEC Certification:**
- **Safety:** IEC standards ensure that products meet minimum safety requirements, protecting users and the environment.
- **Performance:** Certification verifies that products perform as intended and meet established performance criteria.
- **Interoperability:** IEC standards promote compatibility between different products, reducing fragmentation in the market.
- **International Trade:** IEC certification is often required for exporting goods, particularly in the electrical and electronic sectors.

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IEC certification – Example

- IEC 61215, which specifies standards for the design and qualification of terrestrial photovoltaic (PV) modules.
- This certification is crucial for ensuring the safety and reliability of solar panels, particularly in terms of their performance and resistance to environmental factors like temperature, humidity, and mechanical stress.

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ITU (International Telecommunication Union)

- Specialized UN agency to develop **efficient and affordable global telecommunication networks and services.**
- ITU is the **intergovernmental body** responsible for **coordinating the shared global use of the radio spectrum.**
- Promoting international cooperation in **assigning satellite orbits**, improving communication infrastructure in the developing world, and establishing the worldwide standards that foster seamless interconnection.



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Three ITU sectors

ITU has three Sectors:

1. Radiocommunication (ITU-**R**);
2. Telecommunication Standardization (ITU-**T**); and
3. Telecommunication **D**evelopment (ITU-**D**).

Each Sector has its own unique characteristics and activities

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1. Radiocommunication (ITU-R)

- The International Telecommunication Union - Radiocommunication Sector (ITU-R) is a UN agency responsible for establishing regulations and standards for radiocommunication systems.
- It manages the radio-frequency spectrum and satellite orbits, crucial for various wireless services.
- ITU-R also develops recommendations for the efficient use of these resources and organizes conferences like the World Radiocommunication Conference (WRC) to update the Radio Regulations.

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2. Telecommunication Standardization (ITU-T)

- Global organization responsible for developing international standards for information and communication technologies (ICTs).
- ITU-T's primary role is to study technical, operating, and tariff questions related to telecommunications and issue Recommendations to standardize them globally.

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3.Telecommunication **D**evelopment (ITU-**D**)

- Empower all people, especially in developing countries, to use and benefit from ICTs, by promoting **digital inclusion** (ensuring everyone has access to and can effectively use digital technologies like the internet, computers, and smartphones, regardless of their background or abilities) and **leveraging the transformative power of technology** (using technology to create significant positive changes and advancements in various aspects of life, business, or society.) for development.
- This involves assisting member states in developing ICT policies, regulations, and infrastructure, fostering skills development, and ensuring digital accessibility for all.

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EN (European Standards)

- European Standards (EN) are technical standards developed and published by the European Committee for Standardization (CEN), European Committee for Electrotechnical Standardization (CENELEC), and European Telecommunications Standards Institute (ETSI).
- EN standards are technical specifications that define requirements for products, processes, services, or terminology.
- They ensure consistency and interoperability within the European market.
- **Example:**
EN standards cover a wide range of sectors, including construction, electrical equipment, personal protective equipment, and more.



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ASME (American Society of Mechanical Engineers) -Publishes codes and standards for mechanical engineering.

- ASME is one of the oldest standards-developing organizations in **America**.
- Ensure that **equipment and processes** meet demanding safety, performance, and environmental criteria.
- It produces approximately **600 codes** and standards covering many technical areas, such as **fasteners, plumbing fixtures, elevators, pipelines, and power plant systems and components**.



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ASTM International

- ASTM International, formerly known as **American Society for Testing and Materials**, is a standards organization that develops and publishes voluntary consensus technical international standards for a wide range of materials, products, systems and services.



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IEEE (Institute of Electrical and Electronics Engineers)

Develops standards in **electronics, computing, and telecommunications.**

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BIS (Bureau of Indian Standards)

National standards body of India Develops and promotes Indian Standards (IS)

- The Bureau of Indian Standards (BIS) is India's national standards body.
- It is **responsible for developing and implementing standards for various products and services in India.**
- BIS also operates conformity assessment schemes to ensure products meet these standards, **including product certification, registration, and hallmarking.**



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AMISA (African Organisation for Standardisation)

Coordinates standardization across African countries

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7. International Classification for Standards (ICS)

- A hierarchical classification system used to categorize standards by field
- Developed by ISO and IEC
- Helps in systematic organization and search of standards